Chapter 1
Introduction to Electrostatics

Diego Juarez.

These notes are written in the notation used in Jackson: charge density $\rho(\mathbf{x})$, electric field $\mathbf{E}(\mathbf{x})$, scalar potential $\Phi(\mathbf{x})$, outward unit normal $\mathbf{n}$, and permittivity ε_0 in SI units.

Prepared from the material spanning pp. 24–48 of Jackson, Chapter 1.

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Global framework of electrostatics

Electrostatics concerns time-independent charge distributions and the fields they generate. In Jackson's presentation, the subject is built from three layers:

1. the empirical law of force between charges (Coulomb's law),
2. the field description in terms of $\mathbf{E}(\mathbf{x})$ and $\Phi(\mathbf{x})$,
3. the differential and variational formulations of boundary-value problems.

The real power of the subject lies in the transition from forces between individual charges to field equations and then to general methods for solving them under boundary conditions.

1. Coulomb's Law

Coulomb's law is the experimental starting point of electrostatics. For two point charges q_1 and q_2 located at $\mathbf{x}_1$ and $\mathbf{x}_2$, the force on q_1 due to q_2 is

$$\mathbf{F} = \frac{1}{4\pi\varepsilon_0} q_1 q_2 \frac{\mathbf{x}_1 - \mathbf{x}_2}{|\mathbf{x}_1 - \mathbf{x}_2|^3}.$$

Its content is richer than the formula alone suggests. It asserts:

- the force is central, directed along the line joining the charges;
- the magnitude scales as $1/r^2$;
- attraction and repulsion are encoded by the algebraic sign of $q_1 q_2$;
- forces obey linear superposition.

At a graduate level, one should distinguish the physical and mathematical roles of the law. Physically, it is an empirical statement valid for charges at rest. Mathematically, it provides the kernel from which the entire integral theory of electrostatics follows. In particular, because the kernel is central and scales as r^{-2}, the associated potential will scale as r^{-1}, and this is precisely the function whose Laplacian generates a delta distribution.

The superposition principle is indispensable. If there are many charges, the total force is the vector sum of pairwise Coulomb forces. Without superposition, one would not obtain linear field equations. It is this linearity that ultimately makes Green functions, variational methods, and relaxation schemes possible.

2. Electric Field

The electric field is introduced to separate the source configuration from the response of a particular test charge. Jackson defines it operationally as force per unit charge in the limit of a sufficiently

small test charge:

$$\mathbf{F} = q\mathbf{E}.$$

This limiting process matters. A finite test charge would in general disturb the original charge configuration, so the field is not merely "the force divided by charge" in a naive sense; it is the force-to-charge ratio in the idealized limit where the test charge does not modify the sources.

For a point charge q_1 at $\mathbf{x}_1$,

$$\mathbf{E}(\mathbf{x}) = \frac{1}{4\pi\varepsilon_0} q_1 \frac{\mathbf{x} - \mathbf{x}_1}{|\mathbf{x} - \mathbf{x}_1|^3}.$$

For a collection of point charges,

$$\mathbf{E}(\mathbf{x}) = \frac{1}{4\pi\varepsilon_0} \sum_i q_i \frac{\mathbf{x} - \mathbf{x}_i}{|\mathbf{x} - \mathbf{x}_i|^3},$$

and for a continuous distribution,

$$\mathbf{E}(\mathbf{x}) = \frac{1}{4\pi\varepsilon_0} \int \rho(\mathbf{x}') \frac{\mathbf{x} - \mathbf{x}'}{|\mathbf{x} - \mathbf{x}'|^3} \, d^3x'.$$

This transition from discrete to continuous sources is conceptually important. The density $\rho(\mathbf{x})$ is not merely a convenience; it is the macroscopic quantity appropriate when many microscopic charges are present. Jackson emphasizes that the mathematical idealizations of electrostatics are macroscopic constructs. The field is thus a coarse-grained description whose success depends on scale separation between microscopic structure and macroscopic measurement.

At this stage the Dirac delta function naturally appears. A point charge at $\mathbf{x}_i$ is represented by

$$\rho(\mathbf{x}) = q_i \, \delta(\mathbf{x} - \mathbf{x}_i),$$

or more generally,

$$\rho(\mathbf{x}) = \sum_i q_i \delta(\mathbf{x} - \mathbf{x}_i).$$

This is not just notation. It is the exact bridge between localized sources and field equations. The singularity structure of the field near point charges is encoded distributionally through $\delta(\mathbf{x} - \mathbf{x}_i)$.

3. Gauss's Law

Gauss's law gives an integral relation between the electric flux through a closed surface and the enclosed charge:

$$\oint_S \mathbf{E} \cdot \mathbf{n} \, da = \frac{1}{\varepsilon_0} \int_V \rho(\mathbf{x}) \, d^3x.$$

For a point charge, the derivation is geometrical. The quantity $\mathbf{E} \cdot \mathbf{n} \, da$ can be written in terms of the solid angle element $d\Omega$ subtended by the oriented surface element at the charge. One finds

$$\mathbf{E} \cdot \mathbf{n} \, da = \frac{q}{4\pi\varepsilon_0} d\Omega.$$

Integrating over a closed surface yields 4π if the charge lies inside and 0 if it lies outside, giving the point-charge form of Gauss's law. Superposition then extends the result to arbitrary source distributions.

The law is deeper than a computational trick for spheres, cylinders, or planes. It states that the total outward flux is determined only by the total enclosed charge, independent of how that charge is arranged. This is a global constraint on the field. It does *not* determine the full vector field by itself, but it fixes a crucial invariant of the field.

One should also notice the logical ingredients behind Gauss's law: the inverse-square dependence, central forces, and linear superposition. If any one of these failed, the surface integral would no longer collapse to an enclosed-source statement of this form.

4. Differential Form of Gauss's Law

By the divergence theorem,

$$\oint_S \mathbf{A} \cdot \mathbf{n}\, da = \int_V \nabla \cdot \mathbf{A}\, d^3x,$$

so Gauss's law becomes

$$\int_V \left(\nabla \cdot \mathbf{E} - \frac{\rho}{\varepsilon_0} \right) d^3x = 0.$$

Because this holds for arbitrary volumes V, one obtains the local equation

$$\nabla \cdot \mathbf{E} = \frac{\rho}{\varepsilon_0}.$$

This step marks the transition from a global statement to a local field equation. The divergence measures the local rate at which field lines "emerge" from a point, so the equation says that charge density acts as the source density of the electric field. In free space, where $\rho = 0$, the field is divergence-free.

The conceptual shift is major: instead of integrating the effect of sources over all space every time one wants the field, one may instead solve a local partial differential equation. This is the beginning of the field-theoretic viewpoint that dominates the remainder of the book.

5. Another Equation of Electrostatics and the Scalar Potential

The divergence equation alone is insufficient to determine a vector field. A second equation is needed. For electrostatics, the Coulomb integral implies

$$\nabla \times \mathbf{E} = 0.$$

The reason is that

$$\frac{\mathbf{x} - \mathbf{x}'}{|\mathbf{x} - \mathbf{x}'|^3} = -\nabla \left(\frac{1}{|\mathbf{x} - \mathbf{x}'|} \right),$$

so the electric field is the gradient of an integral over a scalar kernel. Since the curl of a gradient vanishes, electrostatic fields are irrotational.

Hence there exists a scalar potential $\Phi(\mathbf{x})$ such that

$$\mathbf{E} = -\nabla\Phi.$$

The minus sign is physical: the electric field points in the direction of decreasing potential. The potential itself is

$$\Phi(\mathbf{x}) = \frac{1}{4\pi\varepsilon_0}\int \frac{\rho(\mathbf{x}')}{|\mathbf{x}-\mathbf{x}'|}\,d^3x'.$$

The potential is defined only up to an additive constant. This reflects the fact that only potential differences are physically meaningful in electrostatics. Work done in moving a test charge from A to B is

$$W = -q\int_A^B \mathbf{E}\cdot d\mathbf{l} = q(\Phi_B - \Phi_A).$$

Thus $q\Phi$ is potential energy, and the line integral of $\mathbf{E}$ depends only on the endpoints. In particular,

$$\oint_C \mathbf{E}\cdot d\mathbf{l} = 0.$$

This path independence is the defining signature of conservative fields.

6. Surface Distributions of Charges and Dipoles; Discontinuities in Field and Potential

A surface-charge density $\sigma(\mathbf{x})$ concentrated on a surface S produces a jump in the normal component of the field. If $\mathbf{E}_1$ and $\mathbf{E}_2$ are the limiting fields on the two sides of the surface and $\mathbf{n}$ points from side 1 to side 2, then

$$(\mathbf{E}_2 - \mathbf{E}_1)\cdot\mathbf{n} = \frac{\sigma}{\varepsilon_0}.$$

The tangential component remains continuous in electrostatics because $\nabla\times\mathbf{E} = 0$ implies the vanishing of the circulation around an infinitesimal loop crossing the boundary.

The scalar potential due to a surface charge is

$$\Phi(\mathbf{x}) = \frac{1}{4\pi\varepsilon_0}\int_S \frac{\sigma(\mathbf{x}')}{|\mathbf{x}-\mathbf{x}'|}\,da'.$$

For ordinary surface charge distributions, Φ is continuous even though $\mathbf{E}$ is not. This is a subtle but fundamental point: a derivative of a continuous function may jump, while the function itself remains continuous.

A dipole layer is more singular. Jackson constructs it as the limit of two oppositely charged nearby

sheets, with surface dipole-moment density $D(\mathbf{x})$ held fixed in the limit. The potential becomes

$$\Phi(\mathbf{x}) = \frac{1}{4\pi\varepsilon_0} \int_S D(\mathbf{x}')\, \mathbf{n} \cdot \nabla' \left(\frac{1}{|\mathbf{x} - \mathbf{x}'|} \right) da' = -\frac{1}{4\pi\varepsilon_0} \int_S D(\mathbf{x}')\, d\Omega.$$

For a dipole layer, the potential itself is discontinuous across the surface:

$$\Phi_2 - \Phi_1 = \frac{D}{\varepsilon_0}.$$

This is one of the most instructive examples in classical field theory. A surface charge creates a discontinuity in the normal derivative of the potential; a dipole layer creates a discontinuity in the potential itself. The hierarchy reflects the distributional order of the source.

7. Poisson and Laplace Equations

Combining

$$\nabla \cdot \mathbf{E} = \frac{\rho}{\varepsilon_0}, \qquad \mathbf{E} = -\nabla\Phi,$$

one obtains

$$\nabla^2\Phi = -\frac{\rho}{\varepsilon_0}.$$

This is Poisson's equation. In charge-free regions,

$$\nabla^2\Phi = 0,$$

which is Laplace's equation.

These equations are central because they replace the nonlocal Coulomb integral by a local partial differential equation. Their mathematical character is elliptic, and that ellipticity is what gives electrostatics its smoothing, averaging, and uniqueness properties.

Jackson also shows how Poisson's equation is satisfied by the Coulomb integral. The key identity is the distributional Laplacian of the Coulomb kernel:

$$\nabla^2 \left(\frac{1}{|\mathbf{x} - \mathbf{x}'|} \right) = -4\pi\delta(\mathbf{x} - \mathbf{x}').$$

This identity should be regarded as the precise analytic statement behind the phrase "the potential of a point charge." Away from the source point, $1/R$ is harmonic; at the source, its singularity carries exactly the delta-function strength needed to represent a localized charge.

8. Green's Theorem

Green's identities follow from the divergence theorem applied to fields of the form $\phi\nabla\psi$. The first identity is

$$\int_V \left(\phi\nabla^2\psi + \nabla\phi\cdot\nabla\psi\right) d^3x = \oint_S \phi\,\frac{\partial\psi}{\partial n}\,da,$$

and the second identity is

$$\int_V \left(\phi\nabla^2\psi - \psi\nabla^2\phi\right) d^3x = \oint_S \left(\phi\frac{\partial\psi}{\partial n} - \psi\frac{\partial\phi}{\partial n}\right) da.$$

Choosing $\phi = \Phi(\mathbf{x}')$ and $\psi = 1/|\mathbf{x}-\mathbf{x}'|$ yields an integral representation of the potential. For $\mathbf{x}$ inside the volume,

$$\Phi(\mathbf{x}) = \frac{1}{4\pi\varepsilon_0}\int_V \frac{\rho(\mathbf{x}')}{|\mathbf{x}-\mathbf{x}'|}\,d^3x' + \frac{1}{4\pi}\oint_S \left[\frac{1}{R}\frac{\partial\Phi}{\partial n'} - \Phi\frac{\partial}{\partial n'}\left(\frac{1}{R}\right)\right] da'.$$

This formula is profound. It states that the potential inside a region is determined by two types of data: the sources within the volume and boundary information on the enclosing surface. Electrostatics is therefore fundamentally a boundary-value theory, not merely a source-integration theory. The surface term contains both Φ and its normal derivative, revealing that one must choose which type of boundary data is physically prescribed.

9. Uniqueness with Dirichlet or Neumann Boundary Conditions

Suppose two solutions Φ_1 and Φ_2 satisfy Poisson's equation in the same volume with the same boundary conditions. Their difference

$$U = \Phi_2 - \Phi_1$$

satisfies Laplace's equation:

$$\nabla^2 U = 0.$$

Using Green's first identity with $\phi = \psi = U$ gives

$$\int_V \left(U\nabla^2 U + |\nabla U|^2\right) d^3x = \oint_S U\frac{\partial U}{\partial n}\,da.$$

Since $\nabla^2 U = 0$,

$$\int_V |\nabla U|^2\,d^3x = \oint_S U\frac{\partial U}{\partial n}\,da.$$

For Dirichlet boundary conditions, $U = 0$ on S, so the right-hand side vanishes. For Neumann boundary conditions, $\partial U/\partial n = 0$ on S, and again the right-hand side vanishes. Thus

$$\int_V |\nabla U|^2\,d^3x = 0,$$

which implies $\nabla U = 0$ everywhere. Therefore U is constant. Under Dirichlet conditions the constant must vanish, so the solution is unique. Under Neumann conditions the solution is unique up to an arbitrary additive constant.

This theorem is not just a mathematical nicety. It guarantees that electrostatic problems are physically well-posed once the appropriate boundary data are given. It also shows why specifying both Φ and $\partial\Phi/\partial n$ on a closed surface is in general an overdetermination: that is the Cauchy problem, which is not the natural electrostatic boundary-value problem.

10. Formal Solution with Green Function

A Green function $G(\mathbf{x}, \mathbf{x}')$ satisfies

$$\nabla^2 G(\mathbf{x}, \mathbf{x}') = -4\pi\delta(\mathbf{x} - \mathbf{x}'),$$

so it differs from the free-space kernel $1/R$ by a harmonic function chosen to enforce the boundary conditions:

$$G(\mathbf{x}, \mathbf{x}') = \frac{1}{|\mathbf{x} - \mathbf{x}'|} + F(\mathbf{x}, \mathbf{x}'), \qquad \nabla^2 F = 0 \text{ in } V.$$

Using Green's theorem, Jackson obtains the general representation

$$\Phi(\mathbf{x}) = \frac{1}{4\pi\varepsilon_0} \int_V \rho(\mathbf{x}') G(\mathbf{x}, \mathbf{x}')\, d^3x' + \frac{1}{4\pi} \oint_S \left[G \frac{\partial \Phi}{\partial n'} - \Phi \frac{\partial G}{\partial n'} \right] da'.$$

The freedom in choosing F is used to eliminate one of the two boundary terms. For Dirichlet problems one chooses G_D such that

$$G_D(\mathbf{x}, \mathbf{x}') = 0 \qquad \text{for } \mathbf{x}' \in S,$$

which gives

$$\Phi(\mathbf{x}) = \frac{1}{4\pi\varepsilon_0} \int_V \rho(\mathbf{x}') G_D(\mathbf{x}, \mathbf{x}')\, d^3x' - \frac{1}{4\pi} \oint_S \Phi(\mathbf{x}') \frac{\partial G_D}{\partial n'}\, da'.$$

For Neumann problems one imposes an appropriate condition on $\partial G_N/\partial n'$ instead.

The physical meaning is elegant. The harmonic correction F may be interpreted as the potential generated by charges external to the region or by induced charges on boundaries. In conductor problems, this is the viewpoint behind the method of images. The Green function is therefore not merely a formal device; it is a source-plus-boundary response function.

11. Electrostatic Potential Energy, Energy Density, and Capacitance

If a charge q_i is brought into a pre-existing potential $\Phi(\mathbf{x}_i)$, the work required is

$$W_i = q_i \Phi(\mathbf{x}_i).$$

Summing over the assembly of charges gives for discrete charges

$$W = \frac{1}{8\pi\varepsilon_0} \sum_i \sum_j \frac{q_i q_j}{|\mathbf{x}_i - \mathbf{x}_j|},$$

with self-terms omitted. For continuous distributions,

$$W = \frac{1}{2} \int \rho(\mathbf{x}) \Phi(\mathbf{x})\, d^3x.$$

Using Poisson's equation and integrating by parts gives the field-energy form

$$W = \frac{\varepsilon_0}{2} \int |\mathbf{E}|^2\, d^3x.$$

Hence the electrostatic energy density is identified as

$$w = \frac{\varepsilon_0}{2} |\mathbf{E}|^2.$$

This expression is conceptually powerful because it relocates the energy from "the interaction of charges" to the field filling space. Field theory here becomes more than a calculational reformulation: it changes where one says the energy resides.

Jackson also explains the subtlety that w is positive definite, whereas the interaction energy of unlike charges is negative. The resolution is that the field-energy expression contains divergent self-energy contributions that are absent from the pairwise interaction energy once self-terms are removed.

For conductors, one writes linear relations between total charges Q_i and potentials V_i:

$$Q_i = \sum_j C_{ij} V_j.$$

The coefficients C_{ii} are capacitances; the off-diagonal coefficients are coefficients of induction. The electrostatic energy becomes

$$W = \frac{1}{2} \sum_i Q_i V_i =$$

rac12$\sum_i \sum_j C_{ij} V_i V_j$. Capacitance is thus not merely a property of an isolated object but of a configuration together with its surrounding conductors and boundary conditions.

12. Variational Approach to the Solution of Laplace and Poisson Equations

Jackson introduces the functional

$$I[\psi] = \frac{1}{2}\int_V \nabla\psi \cdot \nabla\psi \, d^3x - \int_V g\psi \, d^3x.$$

If one varies $\psi \to \psi + \delta\psi$, then to first order

$$\delta I = \int_V \nabla\psi \cdot \nabla(\delta\psi) \, d^3x - \int_V g \, \delta\psi \, d^3x.$$

Using Green's first identity,

$$\delta I = \int_V [-\nabla^2\psi - g]\delta\psi \, d^3x + \oint_S \delta\psi \, \frac{\partial\psi}{\partial n} \, da.$$

If $\delta\psi = 0$ on the boundary (Dirichlet class of variations), stationarity requires

$$\nabla^2\psi = -g.$$

Thus Poisson's equation appears as the Euler–Lagrange equation of the functional. For electrostatics one identifies $g = \rho/\varepsilon_0$ and $\psi = \Phi$.

This viewpoint is valuable for several reasons:

- it proves that the solution is an extremum (indeed a minimum) of an energy-like functional;
- it provides systematic approximation schemes through trial functions;
- it connects electrostatics to the general variational structure of mathematical physics.

The method is not merely formal. One chooses a family of trial functions satisfying the boundary conditions, inserts them into $I[\psi]$, and minimizes with respect to the free parameters. The quality of the approximation then depends on how well the trial family captures the true qualitative behavior of the solution. This is why symmetry, regularity at the origin, and correct boundary behavior matter so much in the choice of ansatz.

For Neumann boundary conditions, the functional is modified by a boundary term so that arbitrary variations are compatible with prescribed normal derivative on the surface.

13. Relaxation Method for Two-Dimensional Electrostatic Problems

The relaxation method is the numerical incarnation of the variational and elliptic character of Laplace's equation. On a square lattice with spacing h, one approximates the potential at a grid point (i, j) by ψ_{ij}. Minimization of the discrete analogue of the Dirichlet functional leads to the

update rule

$$\psi_0 = \frac{1}{4}(\psi_N + \psi_E + \psi_S + \psi_W).$$

That is, at equilibrium each interior lattice value equals the average of its nearest neighbors. This is the discrete mean-value property of harmonic functions.

Iteratively replacing old values by neighbor averages drives the discrete configuration toward the minimum of the discretized functional and therefore toward the numerical solution of Laplace's equation. The method is conceptually simple because it mirrors the physics: in the absence of sources, the potential at a point is determined by the surrounding values.

Jackson further notes that better weighted averages can be formed using both cross and diagonal points, improving accuracy by canceling lower-order discretization error. This is the numerical-analysis side of electrostatics: the same structure that yields uniqueness and smoothing in the continuum yields stable iterative schemes on a grid.

Closing synthesis

The sequence of topics in Jackson is tightly organized. Coulomb's law introduces the source-field relation; Gauss's law and $\nabla \times \mathbf{E} = 0$ convert that relation into local equations; the potential reformulates the vector problem as a scalar one; Poisson and Laplace equations define the core elliptic PDE structure; Green's identities and Green functions turn boundary data into integral representations; uniqueness theorems justify the physical meaning of the resulting solutions; energy and variational principles reveal deeper structure; and relaxation methods show how the same mathematics becomes a practical computational algorithm.

In this sense, Chapter 1 is not merely a review of electrostatics. It is the template for much of classical field theory.